

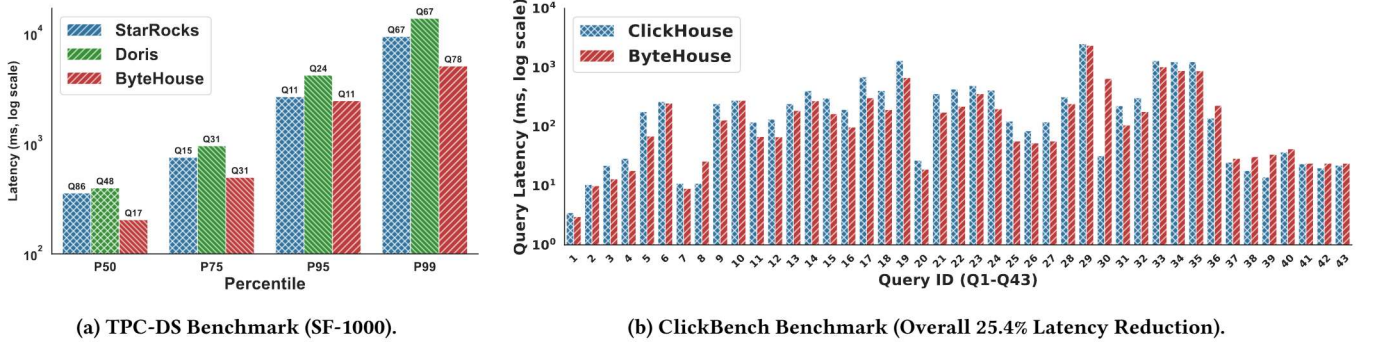


Figure 6: Comparative Results on Standard Benchmarks.

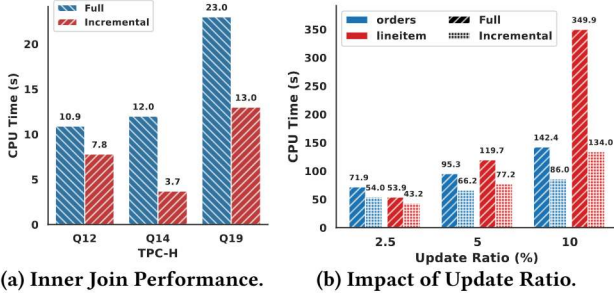


Figure 7: Efficiency of Incremental Processing.

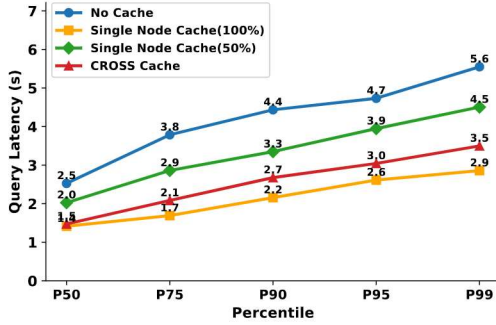


Figure 8: Latency Evaluation of CrossCache.

events. The cluster consists of 180 compute nodes, each equipped with 60 CPU cores (AMD EPYC 9Y24 96-Core Processor, 3.69 GHz), 256 GB of memory, and a 2.5 TB SSD. CrossCache is deployed with one Cache Coordinator and 47 Cache Nodes, where each node provides 2.25 TB of local SSD cache. The evaluated workload comprises the top 1000 historical queries with the largest data scan volumes. Each query completes within 5–7 s when served from cache, while full execution without caching typically takes around one minute. To isolate the effect of distributed caching, we evaluate CrossCache and the single-node cache baselines in a fully isolated environment and enable only one caching mode per run to avoid cross-interference.

Figure 8 reports end-to-end query latency percentiles (P50–P99) across four mutually exclusive settings: no cache, a single-node cache with a 100% hit ratio, the same single-node cache with a 50% hit ratio, and CrossCache. The cache hit ratio is controlled by adjusting the available cache capacity. Across all configurations, caching substantially reduces query latency. Compared with the single-node cache at a 50% hit ratio, CrossCache reduces latency

across all percentiles, with improvements of about 25% at P50, 18% at P90, and 22% at P99. While CrossCache exhibits marginally higher latency than the idealized single-node cache with a perfect (100%) hit ratio, it consistently achieves lower latency than the single-node cache at a 50% hit ratio. This advantage is particularly meaningful in production environments, where maintaining perfect cache locality on a single node is rarely achievable.

7.4 Effect of AI-driven Optimization

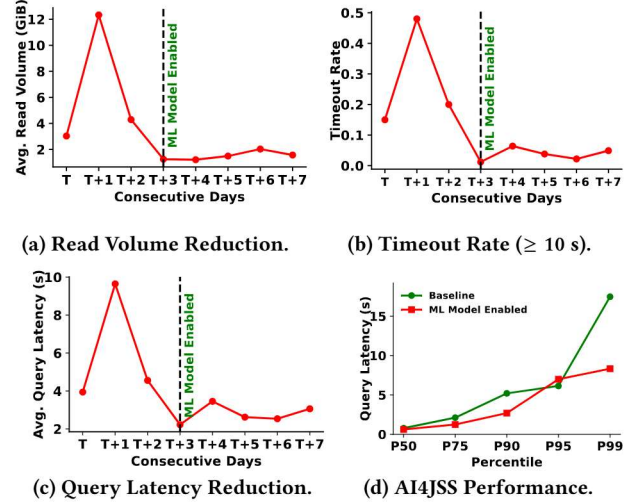
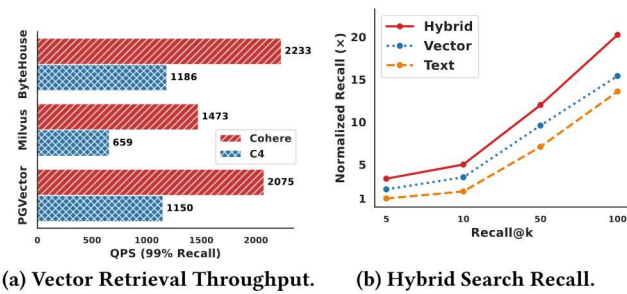


Figure 9: Efficiency of AI-driven Optimization.

Figure 9 illustrates the performance of two AI-driven approaches on query execution efficiency. The experiment was conducted on a production cluster consisting of 100 nodes, each equipped with 192 CPU cores and 1.48 TiB of memory. Figures 9a-9c present the results of the Predicate Pushdown Selectivity (PPS) experiment, where an ML model predicts filter selectivity and dynamically adjusts predicate pushdown strategies. After enabling the model (day T+3), the system exhibits notable reductions in both I/O and query latency: the average read volume drops by up to 87%, the proportion of queries exceeding 10 seconds (timeout rate) drops sharply, and the average query latency is reduced accordingly. Figure 9d further evaluates the Join Side Selection (JSS) experiment, where an ML model is employed to optimize build/probe side selection.

The experiment evaluates 1,000 join queries sampled from real production traces, comparing the latency distributions of the proposed approach against the baseline optimizer. Across all percentiles, the AI-driven approach achieves 15%–45% lower query latency, with particularly strong improvements at the tail (P95–P99) where long-running queries dominate overall cost. Across all latency percentiles (P50–P99), the ML model consistently outperforms the baseline, with the most significant gains observed in the tail latencies (P95–P99). Overall, these results confirm that AI-driven optimization substantially improves scan and join efficiency, delivering measurable gains in real-world analytical query processing.

7.5 Hybrid Multimodal Query Processing



(a) Vector Retrieval Throughput. (b) Hybrid Search Recall.

Figure 10: Performance of Multimodal Query Processing.

Figure 10 evaluates the multimodal query processing capability of ByteHouse. Figure 10a reports vector retrieval throughput (QPS at 99 % recall) on the Cohere [15] and C4 [5] datasets using the VDBench framework [80], comparing ByteHouse with Milvus [67] and pgvector [47]. The workload consists of hybrid queries that combine vector similarity search with a 1% scalar filter. Across both datasets, ByteHouse achieves the highest throughput, improving performance by 50–60% over Milvus on Cohere and by more than 50% on C4, while also maintaining a clear lead over pgvector. These gains stem from ByteHouse’s multi-layer vector index choices and cross-table runtime filtering.

Figure 10b presents the multimodal retrieval accuracy using the MS MARCO [8], which contains 8.84M passages (we sample 2M for testing). Each passage is represented by the BGE-M3 embedding model [13]. We evaluate three retrieval approaches in ByteHouse (i.e., Vector Search, Text search, and Hybrid Search) under the recall metric at different top- k thresholds. The results shows that Hybrid Search consistently achieves the best overall recall, outperforming the single-modality baselines. At Recall@100, it yields approximately a 30% improvement over vector search and a 50% improvement over text search. These results demonstrate that ByteHouse effectively integrates semantic embedding signals with lexical matching within a unified execution framework.

8 Lessons Learned

Durability and Data Organization. Our experience indicates that storage latency is dominated by the write path, where durability and isolation guarantees must be enforced during ingestion. Write-through policies [16, 65] simplify reliability but often suffer

from high cold-read latency under cache misses, whereas write-back policies [6, 45] hide write latency through buffering and deferred persistence, which increases the complexity of durability semantics. A key lesson across these policies is that durability and availability should be decoupled so that persistence can proceed asynchronously without compromising service continuity. We also believe that hybrid data layouts [50] that keep hot tuples in a row format and colder regions in a compressed columnar format provide a practical way to accommodate both update-intensive operations and bandwidth-efficient analytical scans.

Sustaining High Concurrency. ByteHouse serves analytical workloads that reach $10^4 - 10^5$ read/write QPS, where per-query control overheads, metadata lock contention, and I/O amplification from small fragmented accesses quickly become bottlenecks. Our experience indicates that achieving stable performance at this scale requires coordinated design across planning, compute, and storage, rather than isolated operator-level optimizations. ByteHouse reduces planning-layer control overheads through *prepared statements*, accelerates execution-layer performance for selective workloads (e.g., TopN) via *short-circuit evaluation*, and employs the CrossCache layer to absorb fine-grained accesses and reduce storage-layer I/O amplification. This end-to-end co-design enables scalable throughput with stable tail latency at high concurrency.

Vector Search and Multi-Layer Indexing. A key lesson is that vector computation is becoming a fundamental building block for multimodal analytics. Similarity search workloads are moving beyond simple Top- k retrieval with scalar filters toward richer hybrid patterns, such as similarity-bounded range aggregation [35, 39], vector-driven word-cloud analysis [12, 51, 69], and multi-vector joint recall across embedding spaces [14, 38]. These trends highlight the need for more flexible vector-query semantics and tighter integration of vector search with traditional analytical operators to support increasingly expressive multimodal workloads.

9 Conclusions

This paper presents ByteHouse, a cloud-native shared-storage data warehouse for real-time multimodal analytics. ByteHouse integrates a vertically optimized storage layer with a unified execution framework that supports analytic, batch, and incremental processing. Its fusion-based retrieval operators and hybrid optimization techniques enable efficient multimodal query execution. Evaluations across production and standard benchmarks confirm that ByteHouse is a scalable, high-performance foundation for emerging intelligent data applications.

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